Lab 13 Answers

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May 30, 2026

1 Question 1

1.1 a)

The computation of M on $w = 111$ is

$$q_0 111\bot \;\vdash\; 1q_1 11\bot \;\vdash\; 11q_0 1\bot \;\vdash\; 111q_1\bot \;\vdash\; 111\bot q_{\text{acc}}.$$

Thus the 5×5 tableau is as follows:

q_0	1	1	1	$\bot$
1	q_1	1	1	$\bot$
1	1	q_0	1	$\bot$
1	1	1	q_1	$\bot$
1	1	1	$\bot$	q_{acc}

Table 1: The 5×5 tableau for M on input 111.

1.2 b)

Let $x_{i,j,s}$ be the Boolean variable which is true iff the cell in row i and column j of the tableau contains symbol/state s.

For the input $w = 111$, the starting configuration is $q_0 111\bot$, so

$$\phi_{\text{start}} = x_{1,1,q_0} \wedge x_{1,2,1} \wedge x_{1,3,1} \wedge x_{1,4,1} \wedge x_{1,5,\bot}.$$

The accepting condition is that q_{acc} appears somewhere in the tableau:

$$\phi_{\text{accept}} = \bigvee_{i=1}^{5} \bigvee_{j=1}^{5} x_{i,j,q_{\text{acc}}}.$$

In this tableau, this is satisfied because $x_{5,5,q_{\text{acc}}}$ is true.